

Hillary Oyaro

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Summary

Accomplished Cloud & DevOps Engineer with experience building scalable cloud-native applications, automating infrastructure, and deploying AI-powered systems across AWS and GCP environments. Skilled in Linux administration, Kubernetes, Terraform, CI/CD automation, and machine learning operations, with a strong focus on reliability, security, observability, and high availability. Proven ability to develop secure backend systems, manage production infrastructure, and support AI/LLM platforms through automated deployment pipelines, model deployment workflows, and cloud-native engineering best practices.

Work Experience

DevOps Engineer

May 2024 – Present

UPWORK, SAN FRANCISCO, CALIFORNIA UNITED STATES

- Deploy and migrate monolithic systems to AWS using Kubernetes, maintaining 99.95% uptime and adhering to strict SLA targets for government-adjacent clients.
- Implemented MLOps workflows for AI applications, including containerized model deployment, CI/CD automation for ML pipelines, model monitoring, and scalable inference services using Kubernetes, Docker, and cloud-native tooling.
- Built and optimized secure AWS cloud environments by configuring VPC networks, IAM policies, and SSL/TLS secrets management, enhancing data protection, access control, and system resilience.
- Monitor and troubleshoot infrastructure performance using CloudWatch, Splunk, Prometheus, and Grafana to ensure proactive issue detection and high availability.
- Automate infrastructure provisioning and configuration management using Terraform, reducing manual intervention by 60% and ensuring reproducibility of secure environments.

AI Data Quality Reviewer & Evaluator

April 2026 – Present

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- Review and validate multilingual speech transcription and annotation datasets supporting AI, NLP, and speech recognition systems.
- Perform quality assurance for speaker segmentation, timestamp alignment, and transcription accuracy across AI data pipelines.
- Support AI and LLM training workflows through structured data validation, quality analysis, and compliance with annotation standards.
- Identify data inconsistencies and quality issues to improve AI model reliability, accuracy, and overall performance.

Full-Stack Engineer

December 2024 – Present

EMALII, KENYA

- Implemented DevOps and MLOps workflows including CI/CD automation, containerized deployments, monitoring, and reproducible cloud native environments for scalable application and AI service delivery.
- Designed and deployed a cloud-native e-commerce platform integrating secure payment APIs and AI-powered visual search capabilities using PyTorch, TensorFlow, FastAPI, Hugging Face models, and CSV-based product datasets to enhance product discovery and improve user experience.
- Developed and maintained cloud-native software applications and backend services using REST/JSON APIs,

ensuring scalable, secure, and reliable system performance across production environments

- Automated infrastructure operations and data processing tasks using Python scripts, improving efficiency, API validation, and reducing manual operational overhead.
- Built and tested authentication workflows, file handling systems, and technical validation pipelines to ensure data integrity, security, reproducibility, and reliable AI model inference across distributed cloud systems.

Software Engineer

May 2023 – May 2024

ELLOE AI, SAN MATEO, CALIFORNIA, UNITED STATES

- Designed and deployed LLM-powered AI systems for omnichannel platforms using Vertex AI and Hugging Face models, enabling scalable and secure conversational AI experiences across web and messaging queues.
- Implemented MLOps workflows for AI model deployment, monitoring, versioning, and inference optimization to improve reliability, scalability, and operational efficiency of AI services.
- Collaborated with DevOps teams to deploy, monitor, and maintain high-availability cloud services using Docker, Kubernetes, and CI/CD pipelines across multi-cloud environments.
- Troubleshoot AI agent workflows by translating complex user intents into structured, executable actions, improving response accuracy, consistency, and automation reliability.
- Administered quality assurance and evaluation processes for AI-generated outputs, ensuring technical reproducibility, security alignment, and compliance with production reliability standards.

Education

Bachelor of Science in Mathematics and Computer Science

September 2014 – November 2018

TAITA TAVETA UNIVERSITY, VOI, KENYA

Software Engineering Boot Camp in Software Engineering

February 2022 – April 2023

ALX-AFRICA, NAIROBI, KENYA

Additional Skills

Cloud Infrastructure & Security: AWS, VPC, IAM, GovCloud, Cloud Security, Infrastructure as Code, NIST SP 800-171, CMMC, Compliance Auditing

Systems & Networking: Linux, Unix, TCP/IP, VPN, SSL/TLS, DNS, Kubernetes, EKS, Docker

Monitoring & Automation: Terraform, CloudWatch, Splunk, Grafana, Prometheus, Datadog, Ansible, Jenkins, GitHub Actions

Development & Tools: Python, NodeJS, JavaScript, TypeScript, Django, ExpressJS, MongoDB, PostgreSQL, Bash Scripting

Certifications

- Kubernetes and Cloud Native Essential - Cloud Native Computing Foundation (March 2025)
- Google Cloud Developer Expert - Google (June 2023)
- AWS Certified Cloud Practitioner - AWS (July 2022)

Languages

- English
- Swahili